

HATTIESBURG, MS, USA

WMO: 747590

Lat: 31.265N

Lon: 89.253W

Elev: 148

StdP: 14.62

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 13833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.8	28.6	12.0	10.2	36.9	16.8	12.9	39.2	20.0	54.4	18.3	56.2	3.2	0	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	96.0	75.9	93.9	76.0	92.2	75.8	79.7	89.5	78.8	88.3	78.1	87.5	5.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.5	83.3	76.3	138.0	82.6	75.4	133.8	81.9	43.1	89.3	42.3	88.5	41.5	87.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.8	12.9	DB	19.7	98.5	3.9	3.2	16.9	100.8	14.6	102.6	12.4	104.4	9.6	106.7	(4)
(5)			WB	18.4	81.4	3.9	1.2	15.6	82.2	13.3	82.9	11.1	83.6	8.3	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	49.3	53.3	60.3	66.7	73.9	80.5	82.0	82.0	78.0	68.1	57.3	52.2		
	DBStd	13.83	10.43	9.74	8.73	6.75	5.70	3.31	2.80	3.20	4.47	7.94	8.40	9.83		
	HDD50	353	144	69	18	1	0	0	0	0	0	1	26	94		
	HDD65	1839	495	344	195	60	10	0	0	0	1	62	257	415		
	CDD50	6573	122	162	339	503	742	916	991	992	839	561	245	161		
	CDD65	2583	8	17	50	111	287	466	526	527	390	157	27	17		
	CDH74	24549	20	82	364	1006	2685	4701	5273	5302	3459	1397	200	60		
	CDH80	10539	0	5	53	227	1070	2239	2484	2513	1476	449	22	1		
(14) Wind	WSAvg	4.8	5.6	6.1	6.1	6.0	4.9	4.0	3.6	3.6	3.7	4.1	4.6	5.3		
(15) Precipitation	PrecAvg	61.3	6.3	5.7	6.0	5.0	4.8	4.7	5.5	5.3	4.1	3.3	4.7	5.7		
	PrecMax	87.6	13.7	16.5	13.7	17.2	11.6	15.7	14.3	13.8	14.4	13.7	12.6	14.8		
	PrecMin	37.4	1.0	0.7	1.3	0.6	0.5	0.7	1.6	0.2	0.3	0.0	0.7	1.6		
	PrecStd	10.8	3.1	3.4	2.7	3.6	3.2	3.0	2.6	2.9	3.2	2.6	2.6	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.5	80.5	84.0	88.0	93.4	98.8	98.1	98.6	95.0	91.1	83.0	79.2		
		MCWB	65.3	67.4	66.6	71.3	72.9	74.6	75.2	77.3	74.4	71.4	68.9	69.3		
	2%	DB	73.3	76.7	81.5	84.9	90.6	95.2	95.1	95.5	92.6	87.8	79.3	75.4		
		MCWB	65.0	66.2	66.4	68.7	72.6	74.6	76.4	77.3	74.9	71.8	67.3	68.2		
	5%	DB	70.4	72.9	78.5	82.5	88.5	92.8	93.1	93.2	90.6	85.0	75.8	72.3		
		MCWB	63.5	63.9	64.8	67.8	71.9	75.0	76.6	77.0	74.8	70.5	65.4	66.6		
	10%	DB	66.7	69.9	74.9	80.3	86.3	90.7	91.1	91.2	88.4	82.0	72.9	69.0		
		MCWB	60.9	62.9	63.8	66.4	71.0	74.9	76.6	76.8	74.4	69.1	64.8	64.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	71.1	71.4	74.6	77.1	79.9	80.4	81.3	79.1	77.8	73.2	72.4		
		MCDB	73.0	75.4	77.1	82.2	86.5	88.6	90.1	90.8	87.5	83.8	77.9	76.2		
	2%	WB	67.3	69.1	69.5	72.4	75.5	78.4	79.3	80.0	78.0	76.0	71.1	69.6		
		MCDB	71.1	74.0	75.4	79.5	84.1	87.8	89.7	89.8	86.6	82.5	75.3	73.6		
	5%	WB	65.2	66.8	68.0	71.1	74.5	77.6	78.4	79.1	77.1	74.4	69.0	67.6		
		MCDB	69.4	71.5	74.0	78.1	83.0	87.2	88.3	88.7	85.6	80.9	73.7	71.4		
	10%	WB	62.1	64.0	66.4	69.6	73.3	76.8	77.7	78.2	76.2	72.7	66.5	64.8		
		MCDB	65.9	68.5	72.8	76.3	81.8	86.1	87.3	87.5	84.2	78.7	71.4	68.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	21.9	23.3	23.2	21.8	19.7	18.5	18.9	20.1	23.4	24.9	21.4		
		MCDBR	22.3	23.4	23.9	24.0	23.4	22.4	20.8	20.9	22.1	24.5	26.1	21.4		
		MCWBR	14.6	13.7	10.9	10.4	8.2	6.7	6.5	6.6	7.6	10.3	15.0	14.5		
		MCDBR	17.9	18.3	18.6	19.0	19.0	18.5	18.5	18.4	17.8	18.1	20.8	18.2		
	5% WB	MCWBR	13.4	12.8	10.5	10.4	7.6	6.9	6.7	6.9	6.9	8.8	14.0	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.343	0.368	0.402	0.433	0.458	0.520	0.504	0.449	0.378	0.347	0.332		
	taud		2.526	2.483	2.416	2.332	2.275	2.251	2.099	2.142	2.265	2.440	2.475	2.527		
	Ebn at Noon		286	291	290	283	273	265	248	251	261	275	276	278		
	Edh at Noon		27	31	36	40	43	44	51	48	41	32	29	26		
(44) All-Sky Solar Radiation	RadAvg		904	1081	1439	1790	1959	1952	1866	1776	1615	1381	1042	799		
	RadStd		79	154	118	144	118	148	114	123	115	166	95	79		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+245	+164

Nomenclature: See separate page